

# Carry-Over Lottery Allocation: Practical Incentive-Compatible Drafts

Paper Track: Basketball

Open-source repository: <https://github.com/kvr06-ai/cola-sloan-ssac27>

## Introduction

For decades, the NBA has navigated the tradeoff between helping weak teams via the draft and the tanking that ensues. Carry-Over Lottery Allocation (COLA) solves the dilemma. COLA defines weakness using multi-season playoff results, not single-season records. Thus, tanking need not be rewarded. Draft priority carries over between seasons, being reduced when teams have playoff success or make high draft picks. Over time, more help is directed to weaker teams.

COLA is a framework of mechanisms where each may have different rules on how draft priority changes. COLA's definition includes the NBA's own 3-2-1 lottery as an edge case: minimal carry-over exclusions and no reward for losing.

We prove that, under reasonable assumptions, COLA can be incentive-compatible while helping the weakest team, circumventing Banchio and Munro's SSAC 2020 impossibility result for record-based lotteries. Only one assumption requires justification: teams prefer winning games over manipulating the pool of lottery teams. Others are standard assumptions.

## Methods

The incentive-compatibility proof is standard; novelty comes from the reframe that makes it possible.

We discuss tradeoffs within the COLA framework by identifying tunable dials:

- How does a team's draft priority increase?
- How does the priority decrease?
- How does priority map to lottery odds and draft positions?

ZenGM Basketball-GM, an open-source simulator that plays out every game and roster move of realistic fictional leagues, lets us compare different draft mechanisms using these criteria:

- **Targeted help.** How much help do the highest priority teams receive?
- **Long-term parity.** Can all teams be competitive, as measured by long playoff droughts?
- **Incentive compatibility.** Does the mechanism reward losing?

Simplicity is assessed qualitatively. In the paper, we analyze the impact of violating model assumptions (e.g. a team preferring the lottery over the playoffs), and how to mitigate issues, at a cost of complexity.

## Results

We ran nine mechanisms on the same 30 simulated leagues for 25 seasons each and compare them by paired differences with 95% confidence intervals.

The old NBA lottery rewards losing, which caused tanking. 3-2-1 removes that reward but does not help weak teams. Simple COLA removes it too and directs the most help to the teams that have waited longest (Table 1). All three are level on parity. On simplicity, Simple COLA wins outright: no lottery is needed. Teams are ordered by most years without a playoff series win or a top-3 pick; ties are broken by most wins this season. It is one example in a family that all solve the core dilemma.

| Rule               | Incentive compatibility                                             | Targeted help                                                          |                                                               | Long-term parity                                        |
|--------------------|---------------------------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------|
|                    | Help for a worse record<br><i>places gained, 5th-worst to worst</i> | Help for a longer wait<br><i>places gained, 5th-longest to longest</i> | Longest wait for a series win or top-3 pick<br><i>seasons</i> | Wait for the next series win<br><i>seasons, average</i> |
| Old NBA lottery    | +1.32                                                               | 0                                                                      | 16.4                                                          | 4.65                                                    |
| New lottery, 3-2-1 | -0.63                                                               | 0                                                                      | 16.3                                                          | 4.47                                                    |
| Simple COLA        | -0.94                                                               | 6.3                                                                    | 5.5                                                           | 4.58                                                    |

**Table 1.** Three rules on the three criteria, on the same 30 simulated leagues of 25 seasons (means over leagues). Help counts the draft places one team gains when only its record, or only its wait, changes, so a positive number is an earlier pick. The full paper prices the reward for losing from every starting position, for all nine rules we ran.

## Conclusion

COLA shows that helping the weakest teams and ending tanking are not mutually exclusive. It only looked that way because "weakest" was measured by record. Simple COLA resolves the dilemma without complexity, and the wider family lets a league choose its own balance of targeted help, parity, simplicity, and robustness.